

ATHARVA DIGAMBAR

Fourth Year Undergraduate
Bachelor of Technology in Artificial Intelligence & Data Science
AISSMS Institute of Information Technology

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Degree/Certificate	Institute	Year	CPI/%
Bachelor of Technology	AISSMS IOIT, Pune	2022-Present	8.72/10.0
HSC(XII)	National Science Junior College, Akola	2022	91.0%
SSC(X)	Mount Carmel High School, Akola	2020	91.4%

SCHOLASTIC ACHIEVEMENTS

- Maintained a consistent **strong academic performance** throughout the Bachelor of Technology program.
- Contributed to team-based academic projects, showing effective **teamwork, coordination, and leadership skills**.
- Completed multiple academic **projects and mini-projects**, demonstrating innovation and hands-on engineering skills.

WORK EXPERIENCE

CodSoft Python Developer Intern Mar'25 – July'25
Remote

- Developed Python-based applications integrating **Numpy, Pandas, scikit-learn** and other ML libraries to strengthen data handling, preprocessing, and model-building skills.
- Built automation scripts for **data extraction, cleaning, and feature processing**, enabling smoother ML experimentation pipelines.
- Implemented multiple **mini-projects**, including classification/regression models, simple NLP tasks, and API-driven data utilities to deepen hands-on ML development.
- Applied **OOP principles, modular coding, and exception handling** to write clean, scalable, and reusable code suitable for ML workflows.
- Documented experiments, tracked results, and utilized **Git/GitHub** for version control, collaborative development, and reproducible ML pipelines.

PROGRAMMING ACHIEVEMENTS

- 1st Runner-up – PICT Hackathon DataQuest (Pulzion) among 120+ teams.
- Active participant in CodeChef & Kaggle contests.
- Active participant in Hackathons, Competitions and quizzes.

SKILLS

- Languages:** C, C++, Python,
- AI & ML:** Generative AI, Machine Learning, Deep Learning, NLP, TensorFlow, PyTorch
- LLM Frameworks:** LangChain, LangGraph, RAG
- Web Technologies:** HTML, CSS
- Tools:** Git

RESEARCH PAPER AND PUBLICATIONS

AI-Based Video Chaptering & Learning System (Paper) 2025

- Surveyed ASR, Computer Vision, and LLM-based approaches for video chaptering; emphasized multimodal learning for improved educational accessibility and engagement.

AI-Driven Approaches for Drug Repurposing (Paper) 2025

- Surveyed **ML, DL, GNN, and NLP** techniques for accelerating drug repurposing; analyzed real-world case studies and outlined key challenges and future opportunities.

RELEVANT COURSES

Generative AI	Deep Learning	Data Structures & Algorithms
Big Data Analysis	Natural Language Processing	Pattern Recognition and Applications
Artificial Intelligence for Investment	Cloud Computing	Data Analysis
Machine Learning	Computer Networks	Operating Systems
Machine Learning Operations	Data Analysis	Discrete Mathematics and Statistics

PROJECTS

LungGuardian AI: Smart Lung Health Diagnostic System

- Developed an AI-powered diagnostic tool that analyzes **chest X-rays** using **pre-trained CNN architectures (EfficientNetB0, TensorFlow)**, performing binary classification (Normal vs. Malignant) with **model confidence scoring**.
- Implemented a complete **ML inference pipeline** including image preprocessing, resizing, normalization, and model loading for real-time predictions in the browser setup.
- Integrated **Explainable AI (XAI)** components such as **Grad-CAM heatmaps**, saliency maps, and visualization overlays to highlight infected lung regions and improve model interpretability for medical use.

Intelligent AI-Powered Resume Screening Hiring Assistant

- Built an AI-driven recruitment assistant using NLP models (**spaCy, BERT/DistilBERT**) to parse resumes and extract structured insights including skills, experience, education, and certifications.
- Implemented an AI-driven ranking engine using **semantic similarity, skill-matching algorithms, and contextual embeddings** to score candidates based on job relevance, reducing screening time and bias.
- Designed an end-to-end resume analysis pipeline with text preprocessing, entity recognition, skill extraction, and semantic scoring.

Federated and Continual Learning

- Developed and fine-tuned ML models including **Random Forest, SVM, ResNet50**-based prototypes achieving up to **99.39% accuracy** on classification tasks.
- Applied **feature extraction and dimensionality reduction** using **PCA** to enhance model performance and efficiency.
- Mitigated catastrophic forgetting** in continual learning by integrating **distillation loss** during sequential training, **improving knowledge retention**.

QUERYPILOT

- Built an intelligent RAG-based SQL Copilot using **Groq-hosted LLM**, sentence-transformers, and Pinecone (after testing FAISS and ChromaDB), achieving real-time autocompletion inside **MySQL**.
- Implemented **MLOps pipeline with MLflow** to log acceptance metrics (**95%+ acceptance rate across 20+sessions**), using session memory caching and YAML-driven configuration for reproducible and tunable experimentation.
- Deployed with **Docker/Kubernetes** and **CI/CD pipelines**, while adding monitoring and error handling for scalability.

POSITION OF RESPONSIBILITY

Team Lead Aug'25 – Present

Academic Projects & Hackathons, AISSMS IOIT

- Led a **3–5 member** development team to design implement end-to-end **AI/ML/DL and Generative AI pipelines**, including data preprocessing, model training, evaluation.
- Achieved Managed **task distribution, sprint planning, and version control workflows** (Git), ensuring adherence to timelines and code quality standards.